

Total No. of Questions : 6]

SEAT No. :

P5

[Total No. of Pages : 2

APR - 18/TE/Insem. - 5

T.E. (Civil)

ENVIRONMENTAL ENGINEERING - I

(2012 Course) (Semester - II) (301011)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.
- 2) Figures to the right indicates full marks.
- 3) Draw neat figures wherever necessary.
- 4) Assume suitable data if necessary.
- 5) Use of logarithmic tables, slide rule, Mollier charts, electronics pocket calculator and steam tables are allowed.

Q1) a) Define [5]

- i) Sound Pressure
- ii) Sound Intensity
- iii) Sound Pressure level
- iv) Loudness
- v) Equivalent noise level

b) Explain working principle, advantages and disadvantages of adsorption for gaseous pollutant control. [5]

OR

Q2) a) In a workshop there are six machines. The sound pressure level (SPL) of each machine is given below. [5]

Machine No.	1	2	3	4	5	6
SPL in dB	90	88	85	88	70	65

Find the total SPL in dB when all the machines are in operation simultaneously.

b) Write working principle of cyclone to collect the particulate matter. Draw its schematic sketch and write its applications. [5]

P.T.O.

Q3) a) Enlist the data to be required for any town/city water supply project. **[5]**

b) A city has got the following census data **[5]**

Year	1960	1970	1980	1990	2000	2010
Population	22700	28400	32500	37000	41000	44500

Estimate the population in the year 2040 by geometrical increase method.

OR

Q4) a) Describe the different phases involved in any town/city water supply project. **[5]**

b) Find out the fire demand for the city having population of 2200000 using Freeman's formula and National Board of Fire underwriters Formula. **[5]**

Q5) a) Write the acceptable limit and permissible limit as per BIS(10500:2012) for drinking water for the parameters **[5]**

i) Alkalinity

ii) Hardness

iii) Chloride

iv) Turbidity

v) Calcium

b) What is Stokes law? Explain its applicability for the type of solids and write its limitations. **[5]**

OR

Q6) a) Design a rectangular settling tank to treat 5 MLD of water. The detention time is 3 hours and flow through velocity of 8 cm/min. If the depth of the tank is 3m, find the overflow rate and dimensions of the tank. **[6]**

b) What are the objectives of aeration and draw plan and section of cascade aerator. **[4]**

